

Jiawei Chen

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Only a profound understanding of the underlying physics of turbulence can pave the way for truly effective turbulence modeling

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EDUCATION

Ph.D., Mechanics, Tsinghua University, China	2024
M.S.E., Power Engineering, Harbin Institute of Technology, China	2019
B.S.E., Energy and Power Engineering, Harbin Engineering University, China	2017

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate 2025 – Present

(Department of Aerospace Engineering, Iowa State University)

Research:

- (1) Developed a generative learning framework using Transformer-diffusion modeling to predict unsteady aerodynamics and flow-separation dynamics over iced airfoils.
- (2) Used Large Language Model (LLM) to discover subgrid length scales for DES method.

Project:

- (1) An Integrated Numerical and Experimental Study of Wind-Driven Water Film Flow Dynamics Pertinent to Wind Turbine Icing Phenomena (NSF Award: AWD-029845-00001)

Postdoctoral Research Associate 2024 – 2025

(Department of Mathematics, University of Texas at Arlington)

Research:

- (1) Conducted large eddy simulations and implemented vortex identification metrics to resolve complex near-wall turbulent flows.

Project:

- (1) Development and Application of Liutex and Third Generation of Vortex Definition and Identification (NSF EAGER Award: 2300052, PI: Dr. Chaoqun Liu)

PROFESSIONAL SERVICE

Reviewer for academic journals

- *Physics of Fluids*
- *Aerospace Science and Technology*
- *International Journal of Heat and Fluid Flow*

RESEARCH INTERESTS

- Large Language Model
- Generative AI for fluid dynamics
- Turbulence modeling
- Aerodynamics of ice accretion

MENTORING AND VOLUNTEERING

Postdoctoral Research Mentor 2024 – 2025

(Department of Mathematics, University of Texas at Arlington)

- (1) Mentored Emran Hossen (Ph.D. in Mathematics, graduated Spring 2025) on Computational Fluid Dynamics (CFD) and turbulence modeling for his doctoral dissertation.
- (2) Collaboratively developed a Liutex-based sub-grid model for Large Eddy Simulation (LES), leading to a publication and presentation at the AIAA SciTech 2025 Forum.
- (3) Mentee successfully secured a Postdoctoral Scholar position at Michigan Technological University.

Publications

Peer-Reviewed Journal Articles

- [11] **Chen, J.**, Xiao, M., Zhou, Z., & Zhang, Y. (2026). Numerical investigation of discontinuous ice effects on swept wings. *International Journal of Heat and Fluid Flow* 120, 110391.
- [10] **Chen, J.**, Zhou, Z., Xiao, M., et al. (2026). Flow-separation simulations of continuous and discontinuous ice on swept wings. *Aerospace Science and Technology* 174, 111891.
- [9] **Chen, J.**, Yan, C., & Zhang, Y. (2025). Separation Prediction and Roughness Effect Simulation of Iced Airfoils Based on a Modified k- ω - γ Transition Model. *Aerospace Science and Technology*, 110202.
- [8] Zhou, Z., Xiao, M., **Chen, J.**, et al. (2025). Numerical Investigation of Sweep Effects on Turbulent Flow over Iced Wings. *AIAA Journal*, 1–17.
- [7] **Chen, J.**, Zhang, Y., & Fu, S. (2024). Ice Accretion Simulation Incorporating Roughness Effects and Separation correction through a Transition Model. *Physics of Fluids* 36(4). **(editor's pick)**
- [6] **Chen, J.**, Zhang, Y., & Fu, S. (2024). Numerical study of the effects of discontinuous ice on three-dimensional wings. *Physics of Fluids* 36(7).
- [5] **Chen, J.**, Yang, P., Zhang, Y., & Fu, S. (2024). Aerodynamic prediction and roughness implementation toward ice accretion simulation. *Physics of Fluids* 36(1).
- [4] Yang, P., Li, H., **Chen, J.**, Duan, Y., & Zhang, Y. (2024). Influence of Air Flow Field Computation on an Ice Accretion Simulation of Airfoils. *Aerospace Science and Technology*, 109315.
- [3] **Chen, J.**, Zhang, Y., & Fu, S. (2023). Roughness implementation and convective heat transfer coefficient computation toward ice accretion simulation. *Physics of Fluids* 35(11). **(featured article)**
- [2] **Chen, J.**, Cai, L., & Wang, S. (2020). Numerical Study on Effects of Different Excitation Forms on Forced Film Cooling Characteristics. *Journal of Propulsion Technology* 41(4), 860–867.
- [1] Sun, S., Wang, S., Chen, S., Tao, C., Cai, L., & **Chen, J.** (2019). The impact of various forward sweep angles on the performance of an ultra-high-load low-reaction transonic compressor rotor. *Applied Thermal Engineering* 150, 953–966.

Papers Under Review & Preprints

- [3] **Chen, J.**, Yu, Y., Hossen, E., & Liu, C. (2025). Near-Wall Scaling and Separation Prediction of a Rotation-Based Subgrid-Scale Stress Model. *Journal of Turbulence*, under review. arXiv:2507.20043.
- [2] Li, Z., Fang, L., **Chen, J.**, Sharma, A., & He, P. (2025). Field Inversion Machine Learning for Time-Resolved Unsteady Flows in Airfoil Dynamic Stall. Accepted by *AIAA Journal*.
- [1] Xiao, M., Wang, Y., Rodach, F., Font, B., Kurz, M., Suarez, P., Zhou, D., Alcántara-Ávila, F., Zhu, T., Liu, J., Montala, R., **Chen, J.**, Rabault, J., Lehmkuhl, O., Beck, A., Larsson, J., Vinuesa, R., & Pirozzoli, S. (2025). SmartFlow: A CFD-solver-agnostic deep reinforcement learning framework for computational fluid dynamics on HPC platforms. *Computer Physics Communications*, under review. arXiv:2508.00645. Accepted by *Computer Physics Communications*

Conference Presentations and Proceedings

- [1] Hossen, E., Yu, Y., Liu, C., Chen, J., Liu, Z., Yan, Y., Shah, V., & Alvarez, O. (2025). Liutex-Based Sub-Grid Model for Large Eddy Simulation of Turbulent Flow. *AIAA SCITECH 2025 Forum*, 0073.

TECHNICAL SKILLS

Programming Languages

Matlab, Python, Fortran, C++, LaTeX

Software

OpenFOAM, ANSYS, SolidWorks, Paraview, Tecplot, Linux, Slurm

High-performance computing

- Texas Advanced Computing Center
- Nova cluster